Reactions of Organic Compounds

Reaction types, products, tests and yields

Name _____ Class _____ Date _____

REAGENTS AND DATA

Addition: H_2/Ni · Br_2 · HBr · H_2O/H^+ · Substitution: Cl_2/UV · OH^-(aq) · Elimination: conc. H_2SO_4, heat · Oxidation: acidified $K_2Cr_2O_7$ (orange → green)

Molar masses (g/mol): C_2H_4 28.05 · C_2H_5OH 46.07 · CH_3COOH 60.05

Show units on every quantity in every step.

A. Understanding

1. Define, with one example each: (a) addition (b) substitution (c) elimination (d) oxidation (e) condensation.

B. Reaction types and products

2. Classify each reaction: (a) $CH_2{=}CH_2 + H_2 \longrightarrow CH_3CH_3$ (b) $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$ (c) $CH_3CH_2OH \longrightarrow CH_2{=}CH_2 + H_2O$ (d) ethanol → ethanoic acid (e) ethanoic acid + methanol → methyl ethanoate + water (f) $CH_3CH_2Br + OH^- \longrightarrow CH_3CH_2OH + Br^-$

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3. Name the organic product: (a) but-2-ene + H_2 (Ni) (b) but-2-ene + Br_2 (c) chloroethane + OH^-(aq) (d) propan-1-ol heated under reflux with acidified dichromate (e) butan-2-ol warmed with acidified dichromate

C. Tests and selectivity

4. Describe a chemical test, with the observations, to tell apart: (a) hexane and hex-1-ene (b) propan-1-ol and 2-methylpropan-2-ol.

5. Propan-1-ol can be oxidized to propanal or to propanoic acid. How do the conditions decide which product forms?

6. Propene reacts with HBr. (a) Name the two possible products. (b) Which is the major product, and why?

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D. Calculations

7. Ethanol is made industrially by $CH_2{=}CH_2 + H_2O \longrightarrow CH_3CH_2OH$. In one run, 14.0 kg of ethene gives 21.2 kg of ethanol. Calculate the theoretical yield and the percent yield.

8. 9.21 g of ethanol is completely oxidized to ethanoic acid. What mass of ethanoic acid forms?

E. Challenge

9. Starting only from ethene, water and the reagents in the data box, plan a synthesis of ethyl ethanoate. Give each step, its reagents and conditions, and the type of reaction.

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